

Transfinite Interpolation — Coons Patch

$$\begin{aligned} X(u, v) = & (1 - v) \cdot S(u) + v \cdot N(u) \\ & + (1 - u) \cdot W(v) + u \cdot E(v) \\ & - \left[(1 - u)(1 - v) \cdot c_{00} + u(1 - v) \cdot c_{10} \right. \\ & \left. + (1 - u)v \cdot c_{01} + uv \cdot c_{11} \right] \end{aligned}$$

- **Bilinear interpolation** from 4 boundary curves
- **Guarantees C^0 continuity** at block edges
- **Cell counts solved conforming:** opposite sides match

Code: `tmesh_partition.py:_coons` (lines 881-892) *Lit:* Coons 1967, “Surfaces for Computer-Aided Design”